

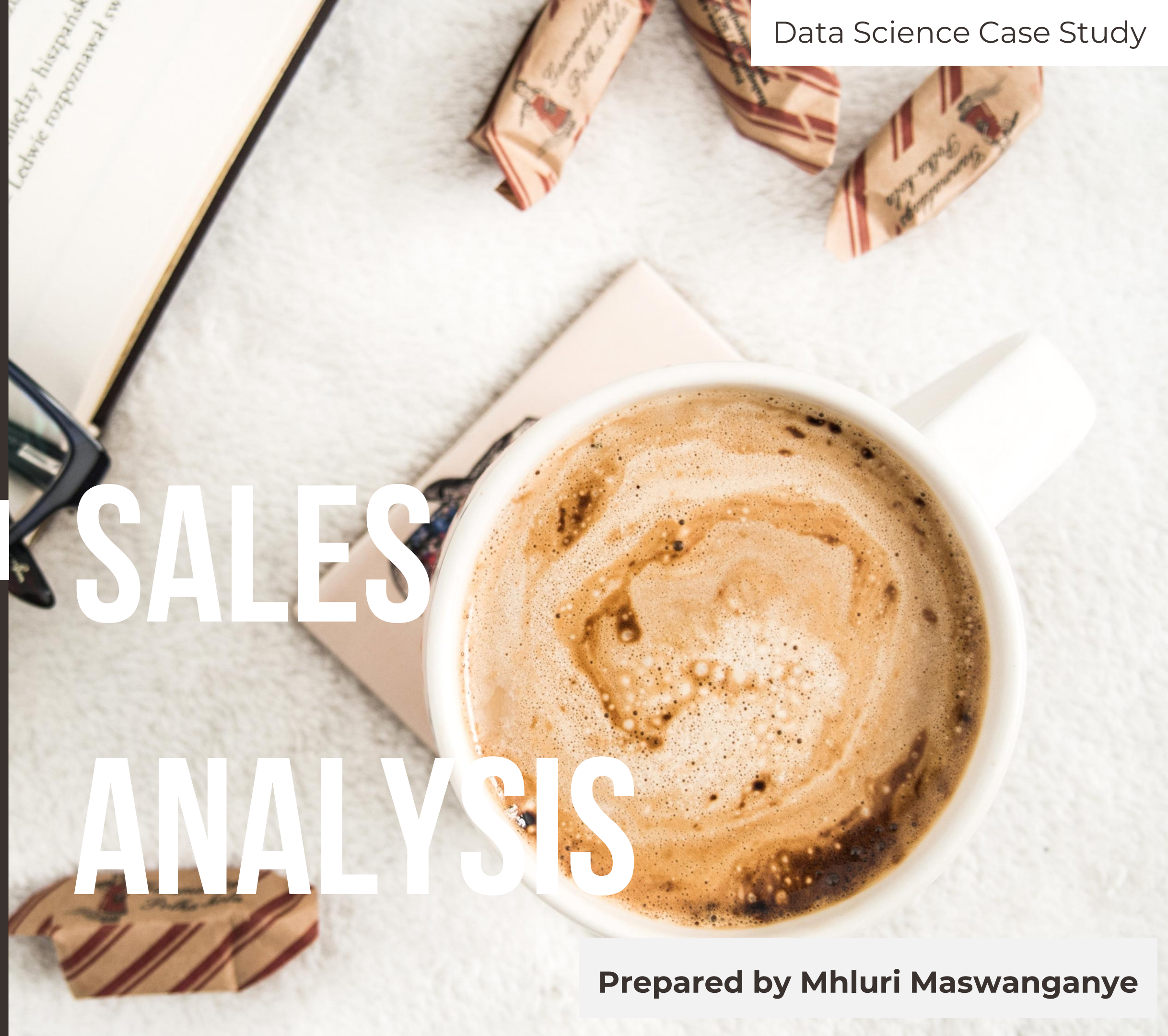
BRIGHT COFFEE SHOP

"Bringing Happiness in Every Cup"

SALES ANALYSIS

Data Science Case Study

Prepared by Mhluri Maswanganye



PROJECT OVERVIEW



Business Problem

Bright Coffee Shop wants to better understand its sales performance in order to make informed business decisions.

Objectives

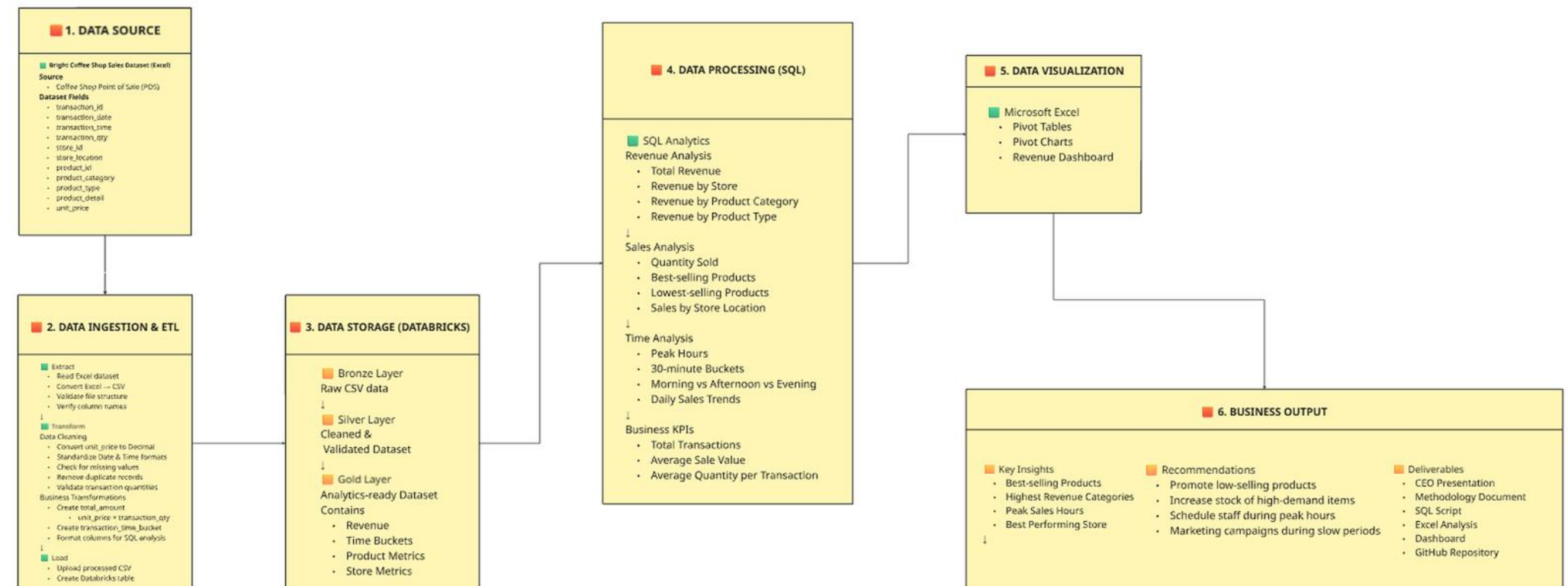
- Analyze sales performance
- Identify best-selling products
- Identify peak sales periods
- Evaluate product categories
- Provide business recommendations

PROJECT PLANNING

PLANNING

Bright Coffee Shop Sales Analysis

Data Flow & Architecture Diagram

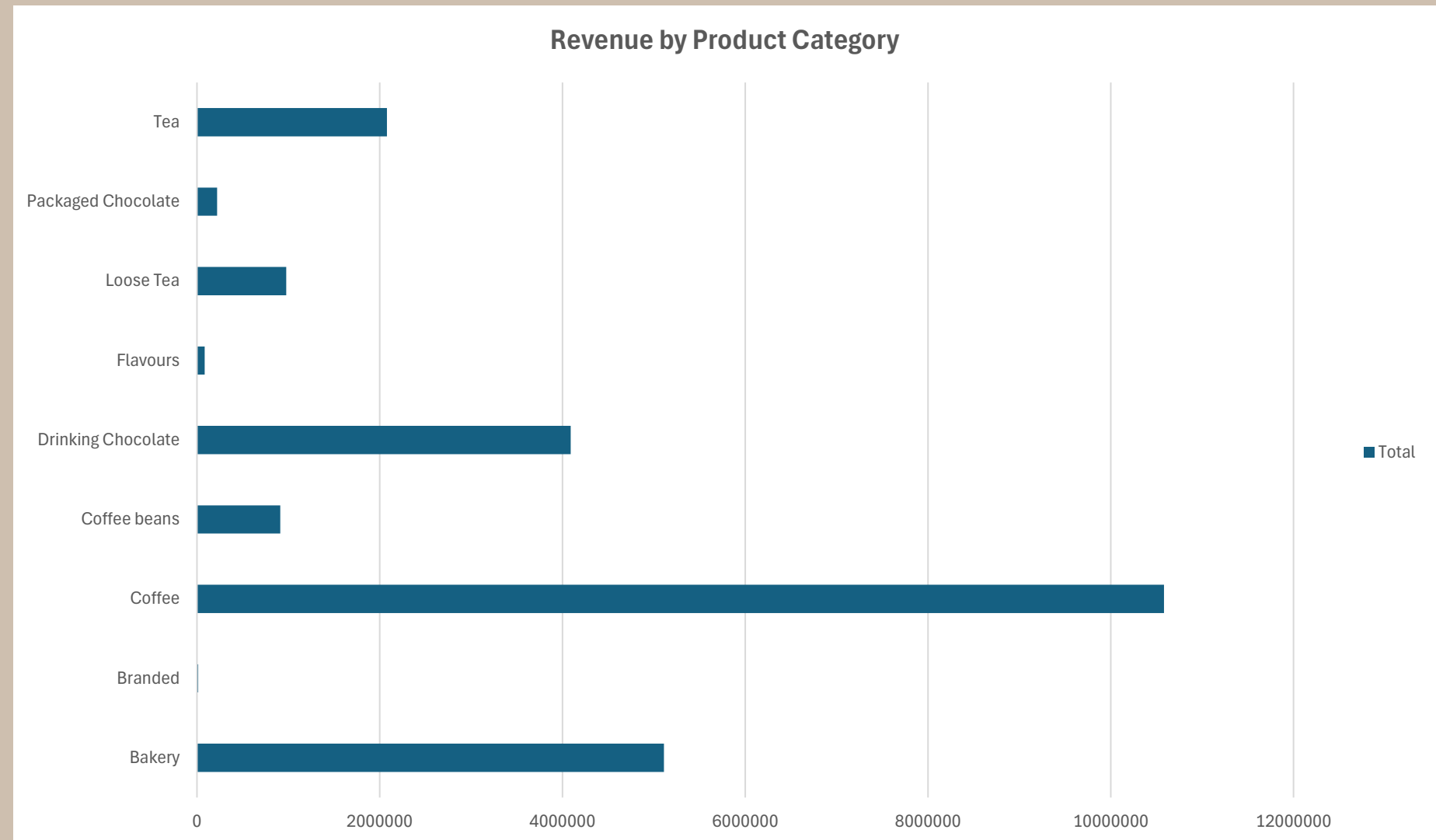


During the planning phase, a data flow and architecture diagram was developed to define the end-to-end analytics process. The workflow begins with the Bright Coffee Shop sales dataset, which is extracted, cleaned, and transformed through an ETL process in Databricks. The processed data is then analyzed using SQL, visualized in Microsoft Excel, and converted into business insights and recommendations to support informed decision-making.

REVENUE BY PRODUCT CATEGORY

Analysis

The analysis shows that Coffee generated the highest revenue among all product categories, making it the primary contributor to overall sales. Bakery products ranked second, indicating that they complement coffee sales and are an important source of revenue. Drinking Chocolate generated moderate revenue, while categories such as Tea, Coffee Beans, Flavours, Loose Tea, Packaged Chocolate, and Branded products contributed comparatively less to total sales.



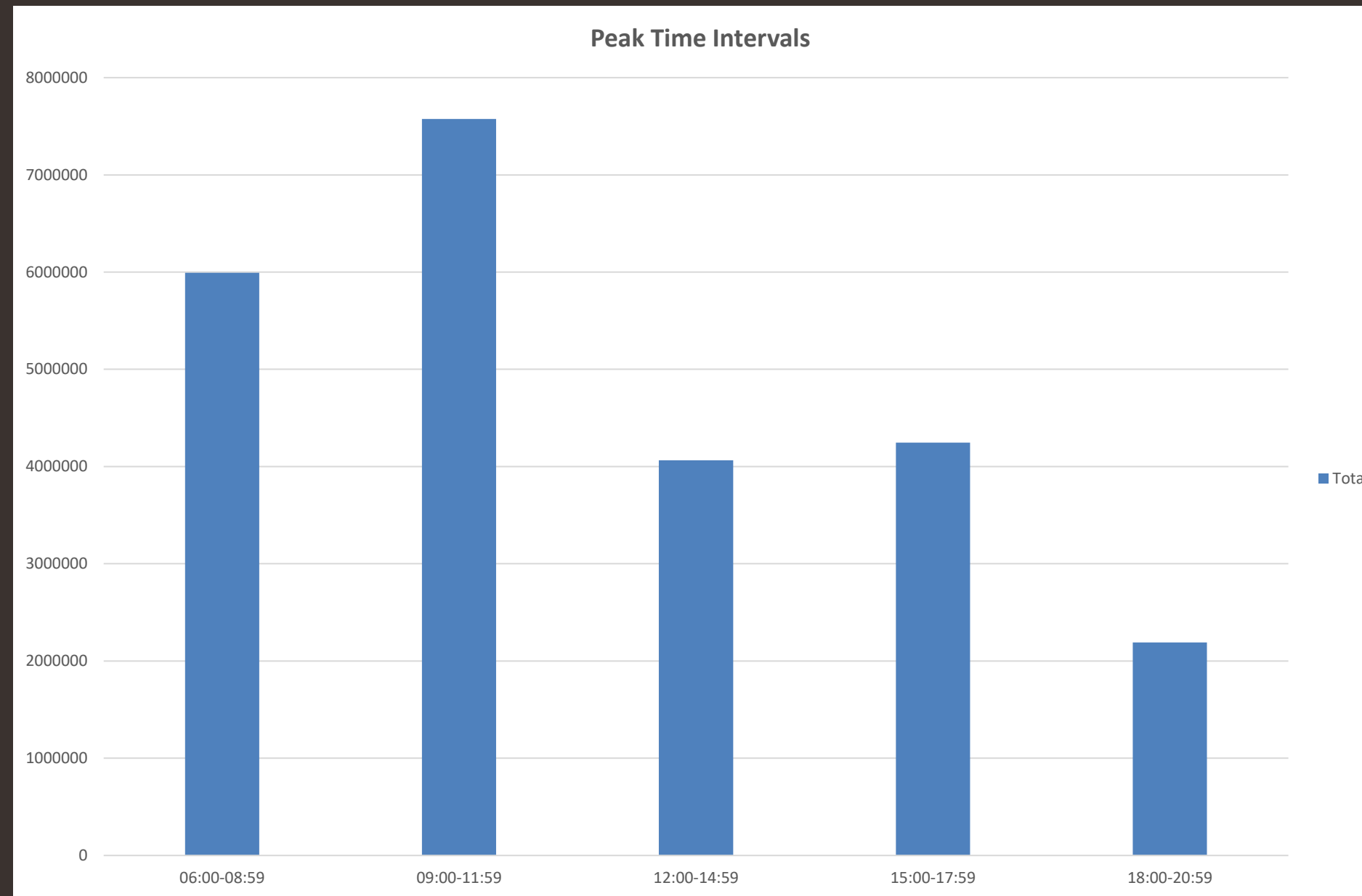
Key Insight

Coffee is the business's strongest revenue driver, suggesting that customer demand is concentrated around coffee-based products.

Recommendation

- Maintain adequate stock levels for coffee products to meet customer demand.
- Continue promoting bakery items alongside coffee through combo offers.
- Introduce targeted promotions or marketing campaigns for lower-performing product categories to improve their sales.

PEAK TIME INTERVALS FOR SALES



Key Insight

Customer demand is highest during the morning hours, particularly between 09:00 and 11:59, suggesting this is the shop's peak operating period.

Analysis

The analysis indicates that the 09:00–11:59 time interval generated the highest sales revenue, making it the busiest trading period of the day. The 06:00–08:59 interval recorded the second-highest sales, showing strong customer activity during the morning. Sales declined during the afternoon, while the 18:00–20:59 interval generated the lowest revenue, indicating reduced customer traffic in the evening.

Recommendation

- Increase staffing levels during the morning rush to improve customer service.
- Ensure sufficient inventory is available before peak trading hours.
- Introduce evening promotions or discounts to encourage sales during slower periods.
- Schedule routine maintenance or staff training during off-peak hours to minimize operational disruption.

BEST-SELLING PRODUCTS

Analysis

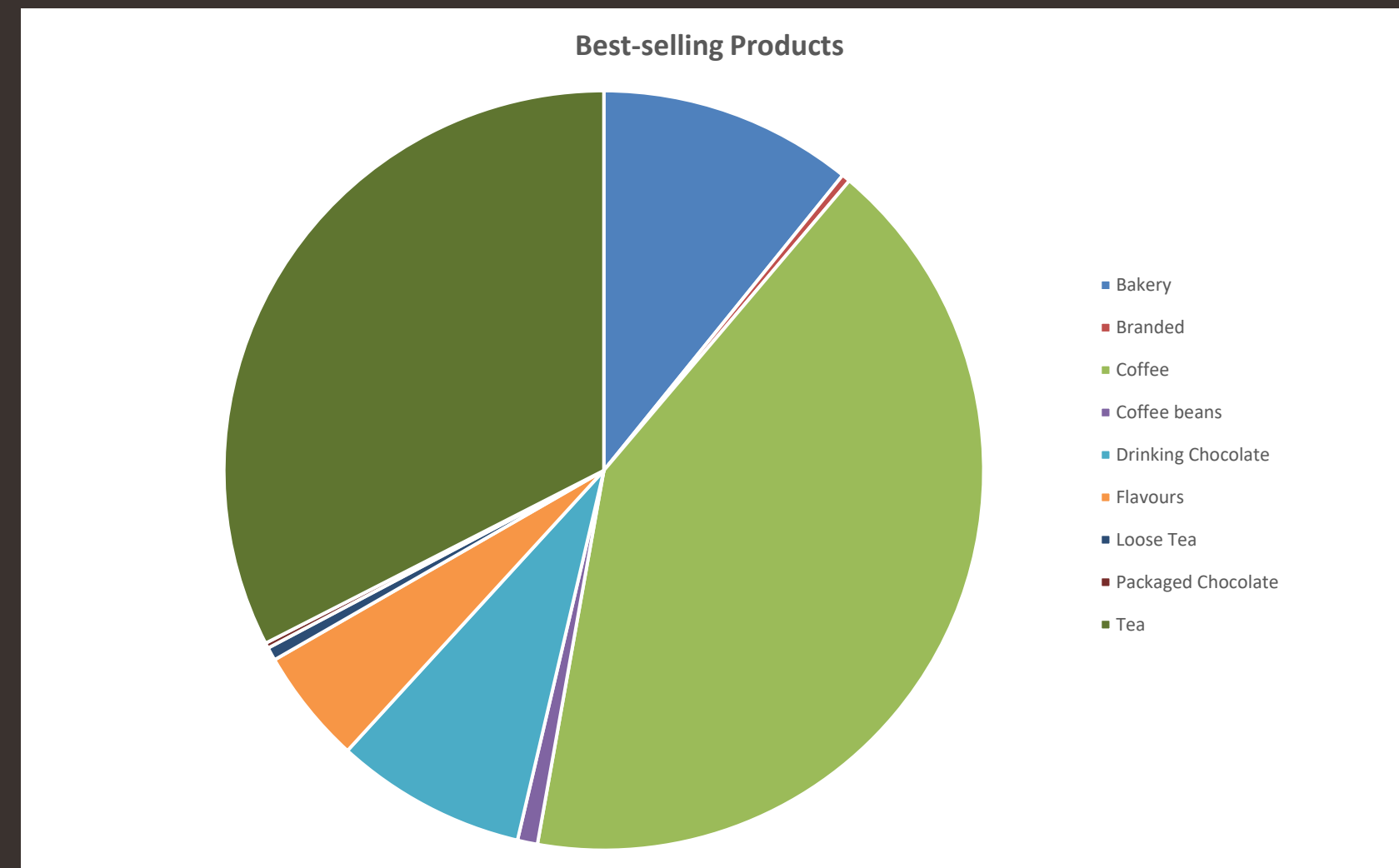
The pie chart shows that Coffee is the best-selling product category, accounting for the largest proportion of items sold. Tea is the second most popular category, followed by Bakery products. The remaining categories, including Drinking Chocolate, Flavours, Coffee Beans, Loose Tea, Packaged Chocolate, and Branded products, contribute a much smaller share of total sales.

Key Insight

Customer demand is strongly concentrated around coffee and tea products, highlighting beverages as the core drivers of sales.

Recommendation

1. Prioritize inventory for coffee and tea products to prevent stock shortages.
2. Continue promoting bakery items alongside beverages through combo offers.
3. Introduce targeted marketing campaigns to improve sales of lower-performing categories.
4. Regularly monitor category performance to respond to changing customer preferences.



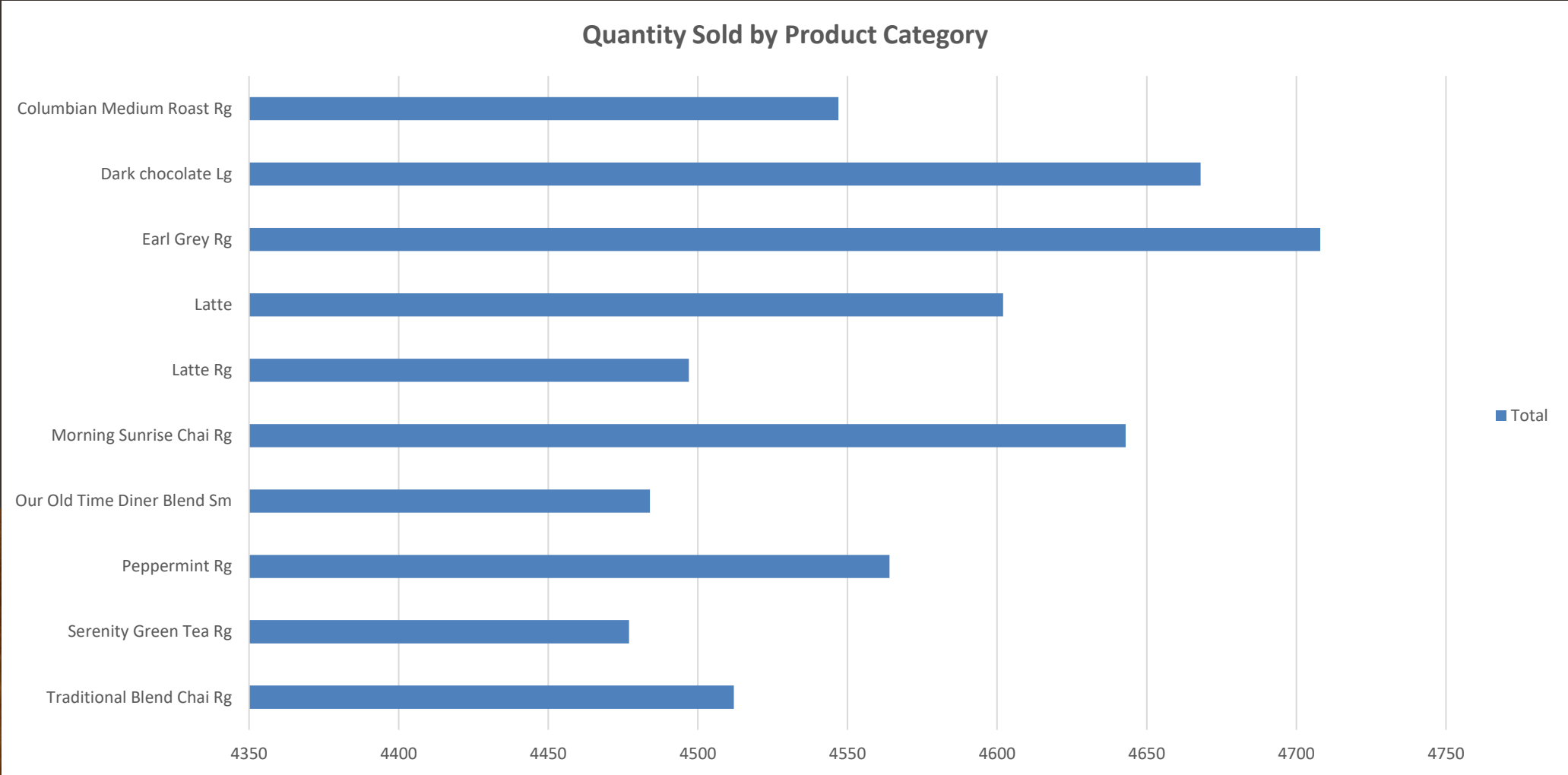


QUANTITY OF ITEMS SOLD BY PRODUCT CATEGORY



ANALYSIS

The analysis highlights the individual products that achieved the highest sales volume. Earl Grey Regular recorded the highest quantity sold, making it the best-selling product. It was closely followed by Dark Chocolate Large and Morning Sunrise Chai Regular, while Latte also demonstrated strong sales performance. Products such as Our Old Time Diner Blend Small, Serenity Green Tea Regular, and Traditional Blend Chai Regular recorded comparatively lower sales volumes.



KEY INSIGHT

A small group of products consistently attracts the highest customer demand, indicating that these items are key contributors to the coffee shop's sales performance.

RECOMMENDATION

1. Maintain sufficient stock levels for the highest-selling products to avoid stock shortages.
2. Feature top-selling products in promotional campaigns and menu highlights.
3. Evaluate lower-performing products to determine whether pricing, marketing, or product placement could improve their sales.
4. Continuously monitor product performance to ensure inventory aligns with customer demand.

Business Insights

The sales analysis revealed that Coffee is the highest-performing product category in terms of revenue and sales volume, making it the core driver of the business. Customer activity is concentrated during the 09:00–11:59 period, indicating that morning hours are the busiest trading window. Additionally, products such as Earl Grey Regular, Dark Chocolate Large, and Morning Sunrise Chai Regular consistently ranked among the best-selling items, while several other products recorded comparatively lower demand.

Business Recommendations

Prioritize Inventory: Maintain adequate stock levels for coffee products and other top-selling items to prevent stock shortages.

Optimize Staffing: Increase staff availability during the morning peak period (09:00–11:59) to improve customer service and reduce waiting times.

Promote Lower-performing Products: Introduce discounts, bundle deals, or marketing campaigns to increase demand for products with lower sales.

Leverage Best Sellers: Feature high-performing products in promotions and seasonal campaigns to maximize revenue.

Monitor Sales Trends: Regularly analyze sales data to identify changes in customer preferences and support data-driven business decisions.

OVERALL BUSINESS INSIGHTS & RECOMMENDATIONS

"We believe that business is not just about profit but also about making a positive impact on the environment and the community."



THANK

Bright Coffee Shop Sales Analysis

Thank you for your time and attention.

I appreciate the opportunity to present the findings and recommendations from this analysis.

🕒 **Presented by**
📍 **Mhluri Maswanganye (Data Scientist)**

🌐 Data Science Case Study

YOU